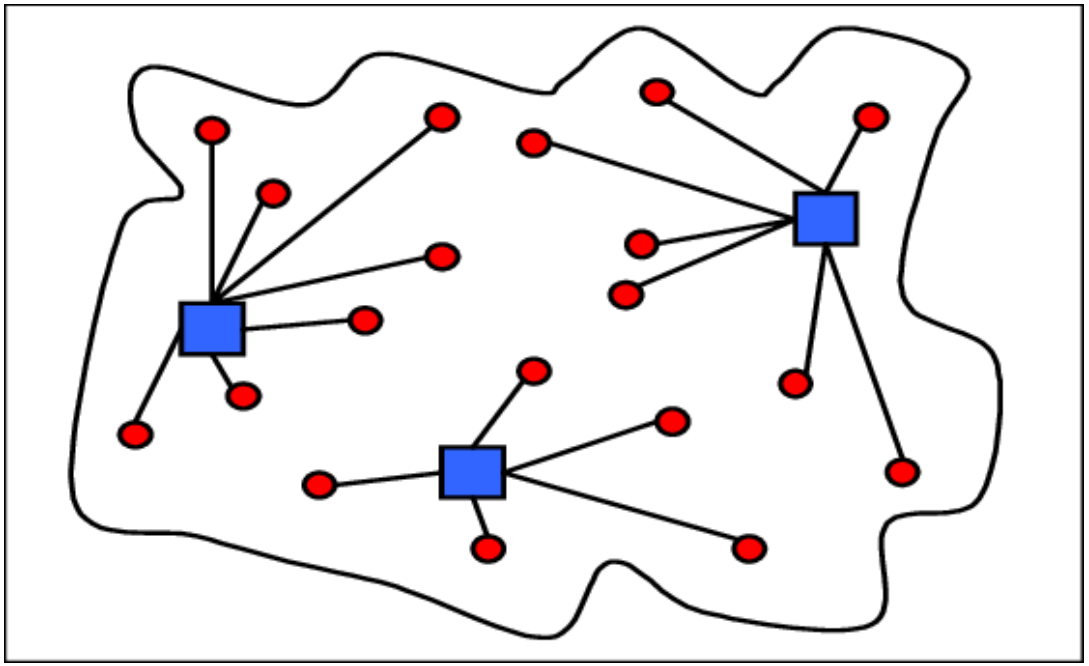
Minimum Norm Matroid Median

Felix Zhou

August 5, 2021

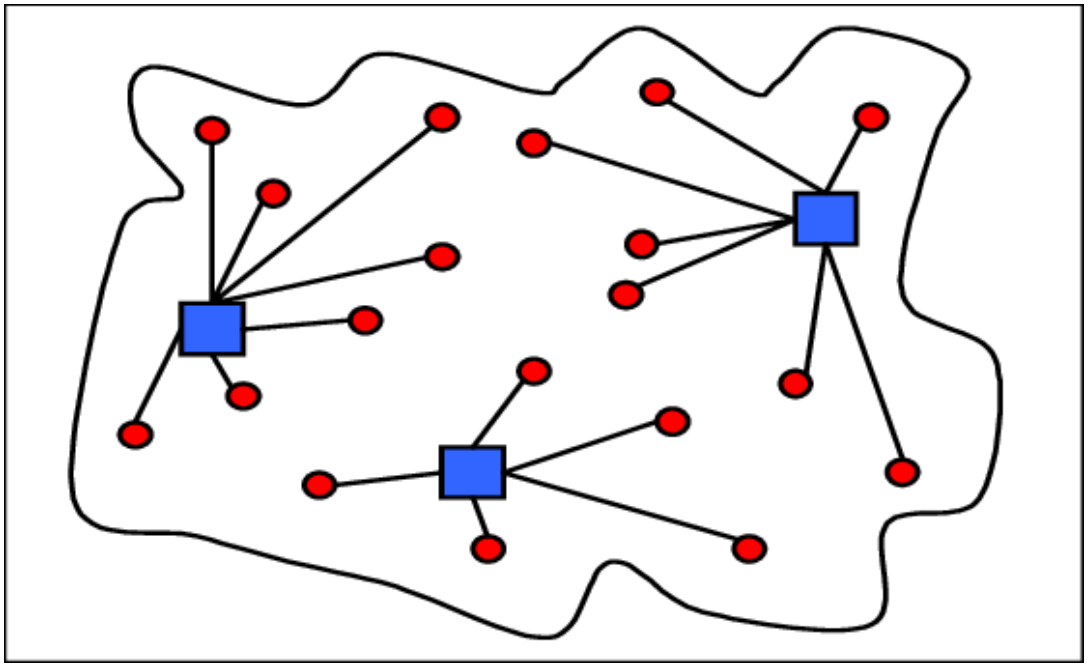
Metric Facility Location Problems

- Models placing bus stops to serve a city



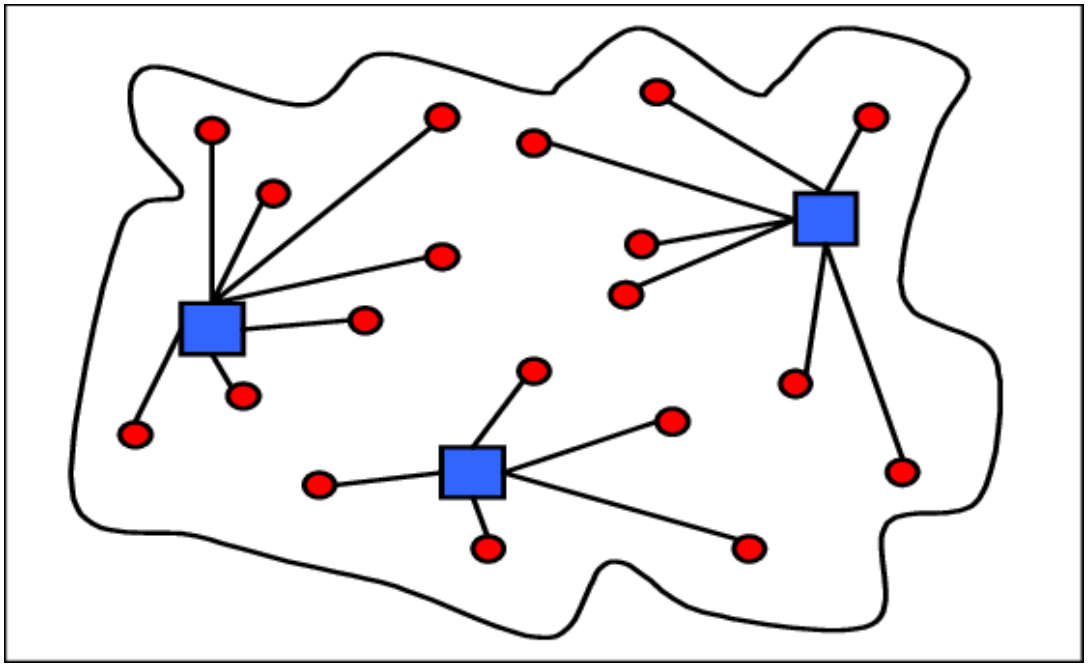
Metric Facility Location Problems

- Models placing bus stops to serve a city
- Opening factories to serve retail outlets



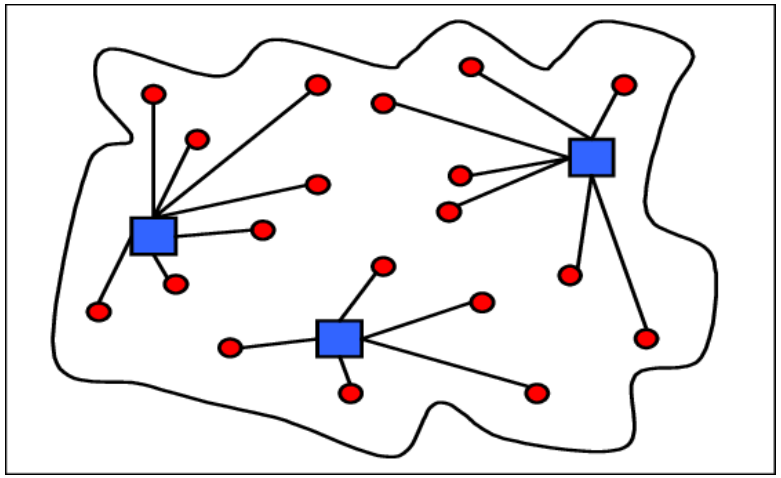
Metric Facility Location Problems

- Models placing bus stops to serve a city
- Opening factories to serve retail outlets
- Clustering problems



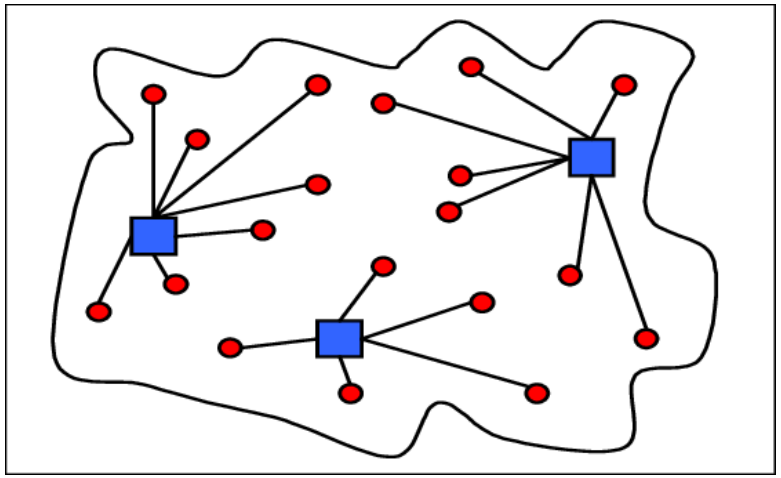
Metric Facility Location Problems

- Finite collection $\mathcal{F}$ of facilities and finite collection $\mathcal{D}$ of clients in a metric space (assume $\mathcal{F} = \mathcal{D}$)



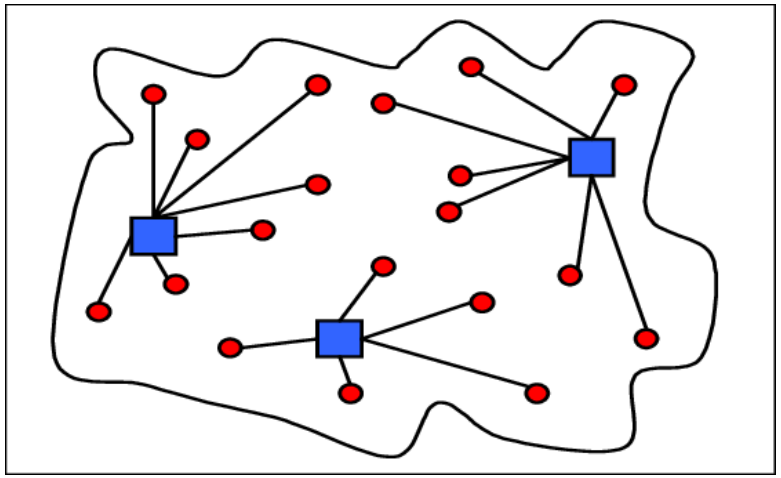
Metric Facility Location Problems

- Finite collection $\mathcal{F}$ of facilities and finite collection $\mathcal{D}$ of clients in a metric space (assume $\mathcal{F} = \mathcal{D}$)
- Open k facilities minimizing the average distance from a client to nearest open facility (k-median) or the maximum distance (k-center)



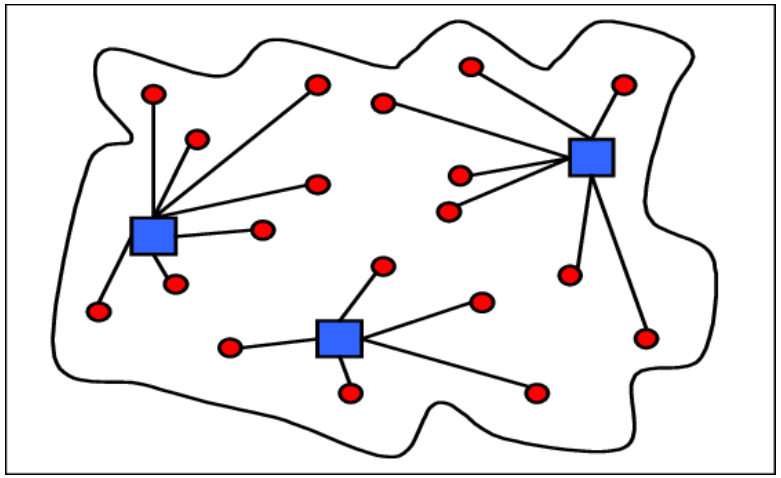
Metric Facility Location Problems

- Finite collection $\mathcal{F}$ of facilities and finite collection $\mathcal{D}$ of clients in a metric space (assume $\mathcal{F} = \mathcal{D}$)
- Open k facilities minimizing the average distance from a client to nearest open facility (k-median) or the maximum distance (k-center)
- Correspond to minimizing the ℓ_1-norm and ℓ_∞ norms of the cost vectors induced by solutions



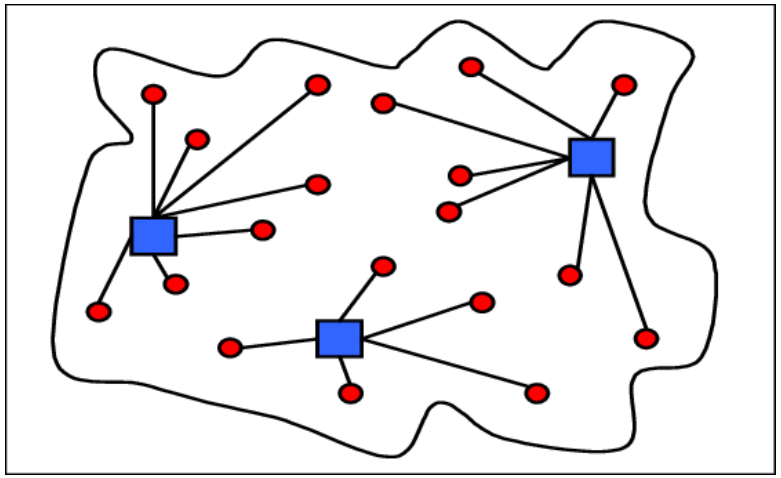
Metric Facility Location Problems

- Minimizing the ℓ_1-norm penalizes a few clients in order to improve the cost of the other clients



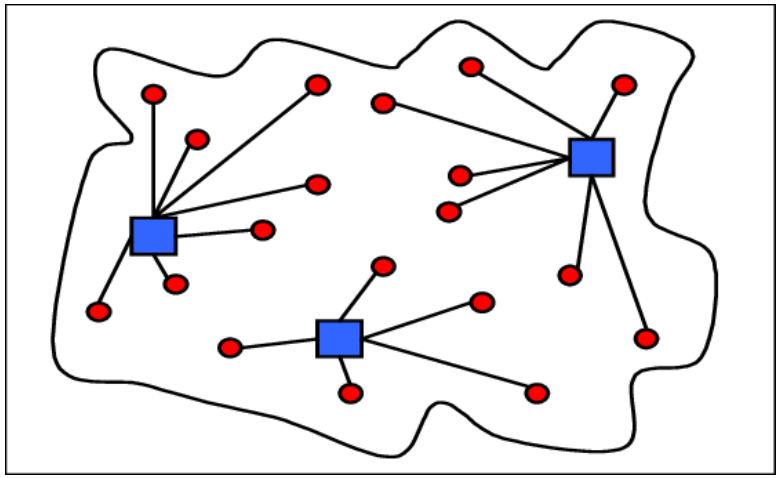
Metric Facility Location Problems

- Minimizing the ℓ_1-norm penalizes a few clients in order to improve the cost of the other clients
- Minimizing the ℓ_∞-norm favours the extreme few at the cost of the majority



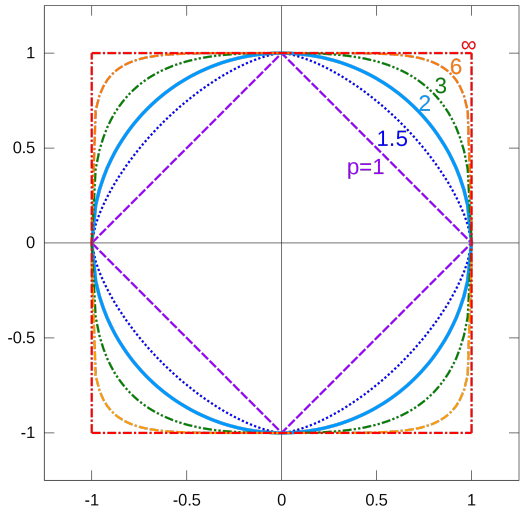
Metric Facility Location Problems

- Minimizing the ℓ_1-norm penalizes a few clients in order to improve the cost of the other clients
- Minimizing the ℓ_∞-norm favours the extreme few at the cost of the majority
- How can we interpolate between these two?



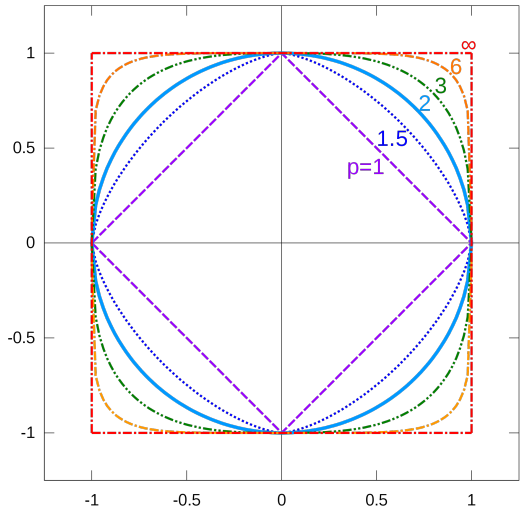
Interpolation

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^{n} x_i^p}$
- $\ell_p \to \ell_\infty$ as $p \to \infty$



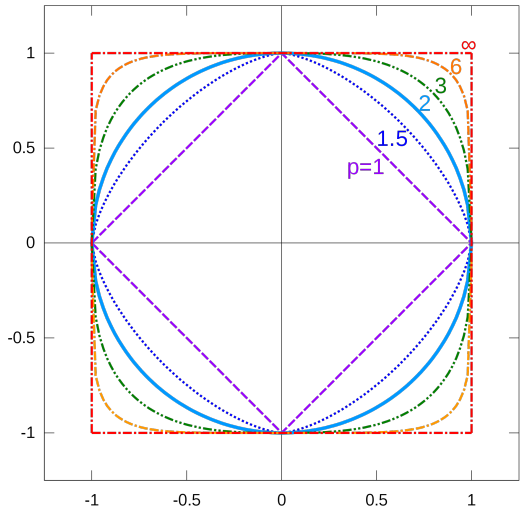
Interpolation

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^{n} x_i^p}$
- $\ell_p \to \ell_\infty$ as $p \to \infty$
- Top-ℓ norms: $\mathrm{Top}_\ell(x)$ is the sum of largest ℓ elements
- $\mathrm{Top}_1 = \ell_\infty, \mathrm{Top}_n = \ell_1$



Interpolation

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^{n} x_i^p}$
- $\ell_p \to \ell_\infty$ as $p \to \infty$
- Top-ℓ norms: $\mathrm{Top}_\ell(x)$ is the sum of largest ℓ elements
- $\mathrm{Top}_1 = \ell_\infty, \mathrm{Top}_n = \ell_1$
- Ordered norms: nonnegative linear combination of Top-ℓ norms



A Generalization

What if we wish to minimize a general monotone, symmetric norm $f : \mathbb{R}_+^n \to \mathbb{R}_+$ over the cost vectors?

- monotone: if $x \leq y$ component-wise, then $f(x) \leq f(y)$

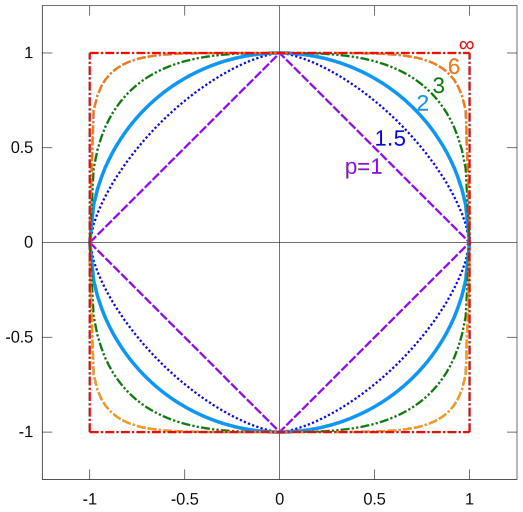
A Generalization

What if we wish to minimize a general monotone, symmetric norm $f : \mathbb{R}_+^n \to \mathbb{R}_+$ over the cost vectors?

- monotone: if $x \leq y$ component-wise, then $f(x) \leq f(y)$
- Write $x^{\downarrow}$ to denote the vector obtained from $x \in \mathbb{R}^n$ by sorting its coordinates in non-increasing order
- symmetric: $f(x) = f(x^{\downarrow})$

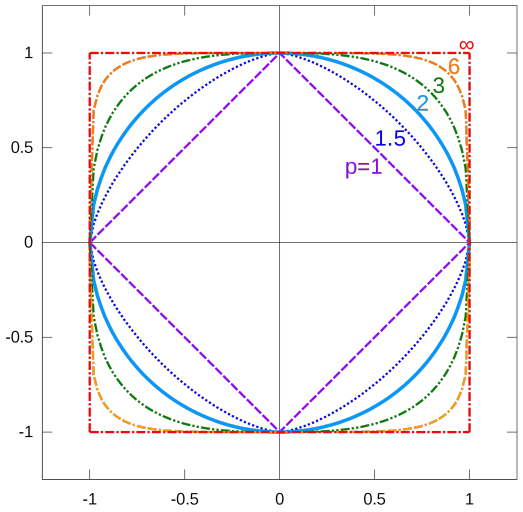
Examples

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^{n} x_i^p}$
- Top-ℓ norms: $\mathrm{Top}_\ell(x)$ is the sum of largest ℓ elements
- Ordered norms: nonnegative linear combination of Top-ℓ norms



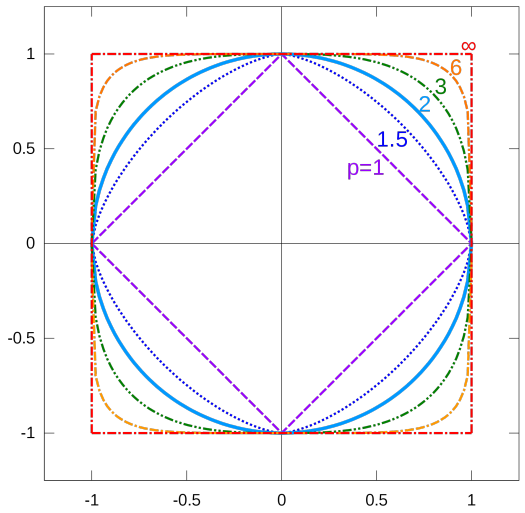
Examples

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^{n} x_i^p}$
- Top-ℓ norms: $\mathrm{Top}_\ell(x)$ is the sum of largest ℓ elements
- Ordered norms: nonnegative linear combination of Top-ℓ norms
- max over a finite collection of monotone, symmetric norms



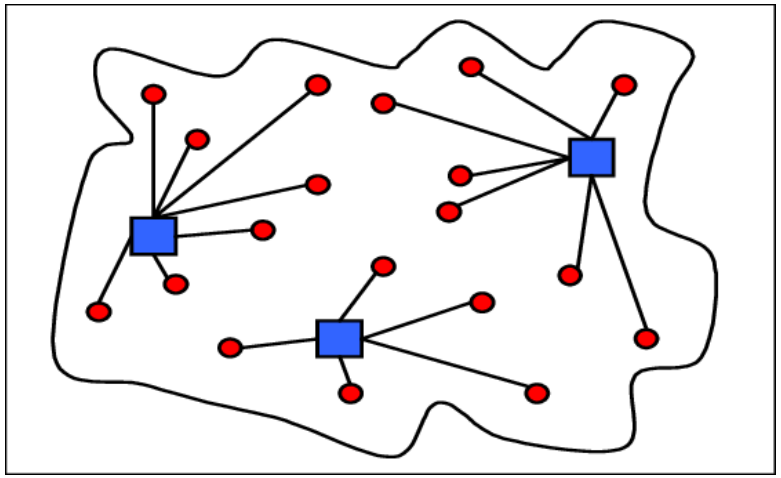
Examples

- ℓ_p norms: $f(x) = \sqrt[p]{\sum_{i=1}^n x_i^p}$
- Top-ℓ norms: $\mathrm{Top}_\ell(x)$ is the sum of largest ℓ elements
- Ordered norms: nonnegative linear combination of Top-ℓ norms
- max over a finite collection of monotone, symmetric norms
- positive linear combination of monotone, symmetric norms



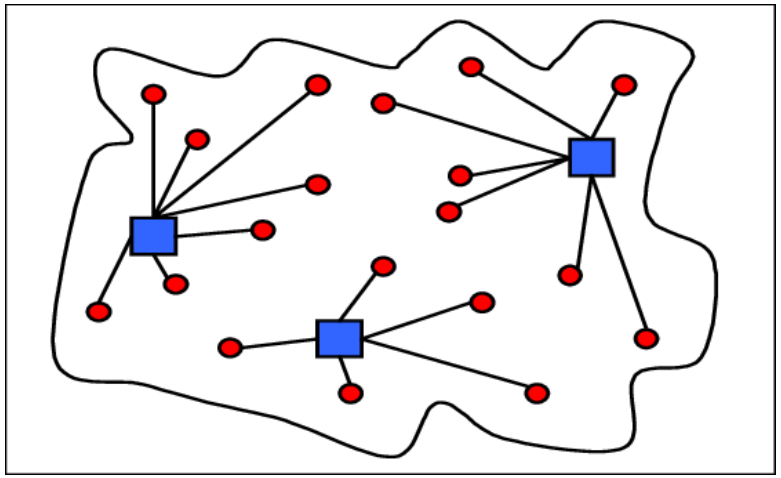
Minimum Norm Matroid Median

- Generalize objective function of k-medians / k-center



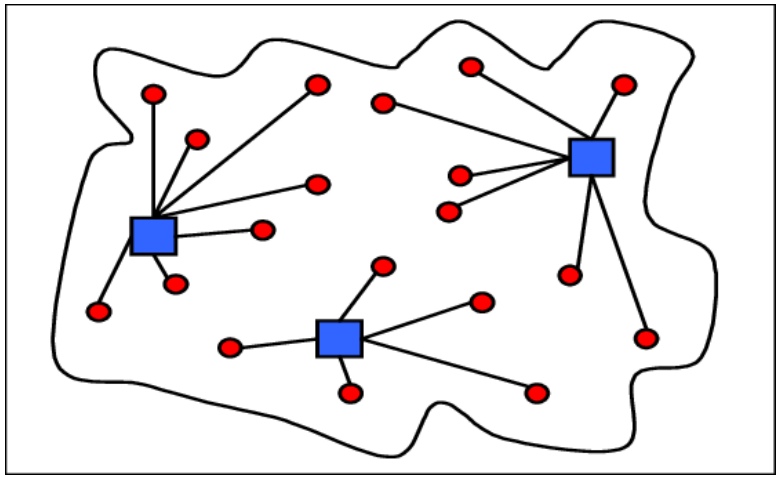
Minimum Norm Matroid Median

- Generalize objective function of k-medians / k-center
- Generalize the "$\leq k$ facilities" to a matroid constraint



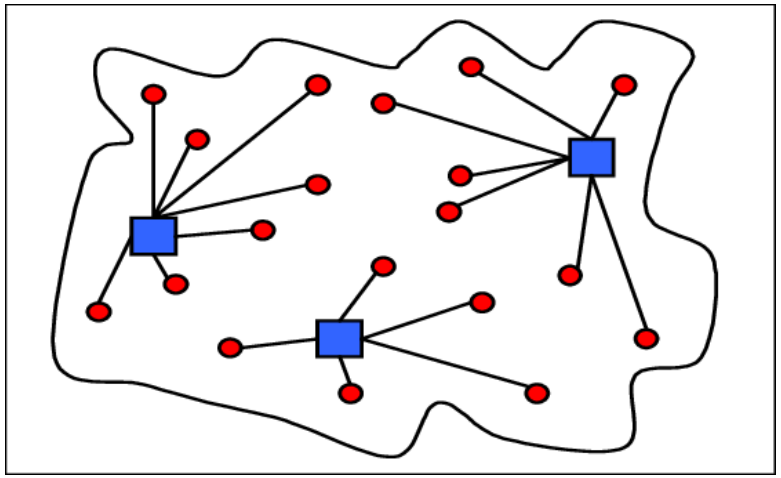
Minimum Norm Matroid Median

- Generalize objective function of k-medians / k-center
- Generalize the "$\leq k$ facilities" to a matroid constraint
- Matroids capture other facility location problems



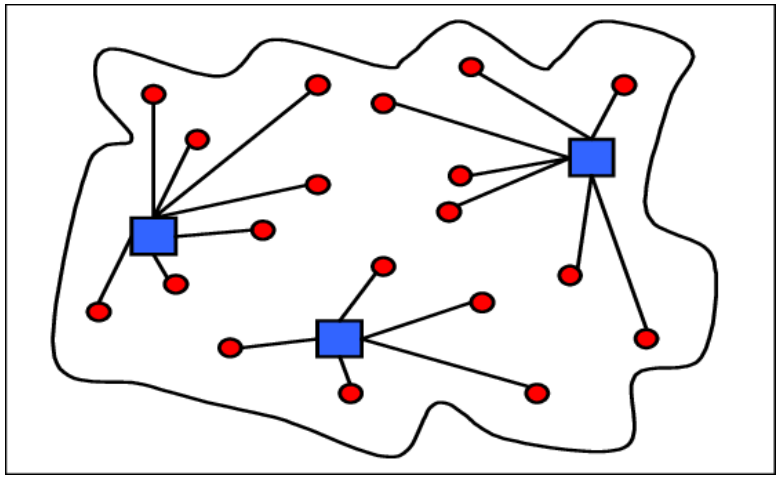
Minimum Norm Matroid Median

- Generalize objective function of k-medians / k-center
- Generalize the "$\leq k$ facilities" to a matroid constraint
- Matroids capture other facility location problems
- data placement, mobile facility location, k-median forest, metric uniform minimum-latency UFL



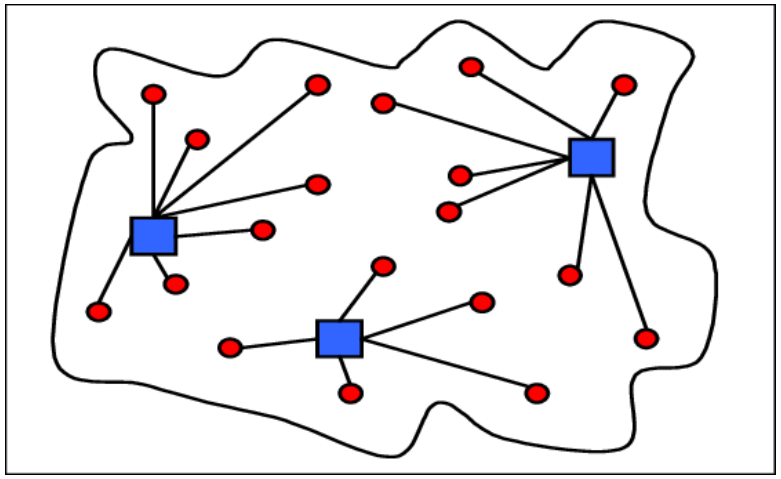
Minimum Norm Matroid Median

- Finite set $\mathcal{F} = \mathcal{D}$ of facilities and clients in a metric space



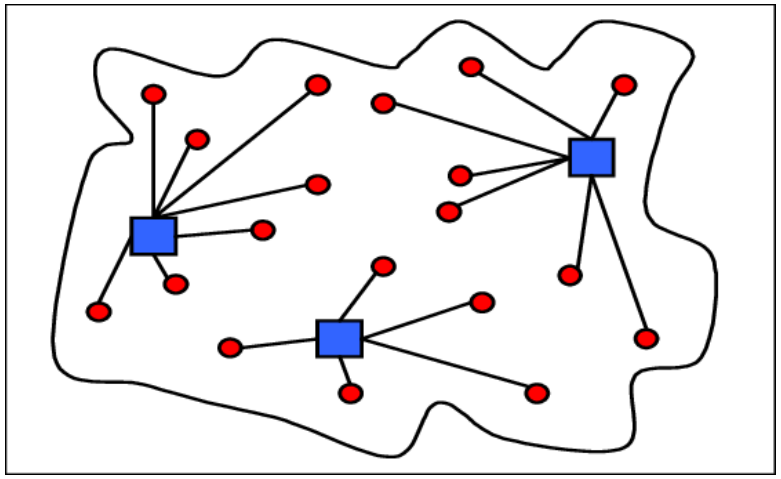
Minimum Norm Matroid Median

- Finite set $\mathcal{F} = \mathcal{D}$ of facilities and clients in a metric space
- A matroid $M = (\mathcal{F}, \mathcal{I})$



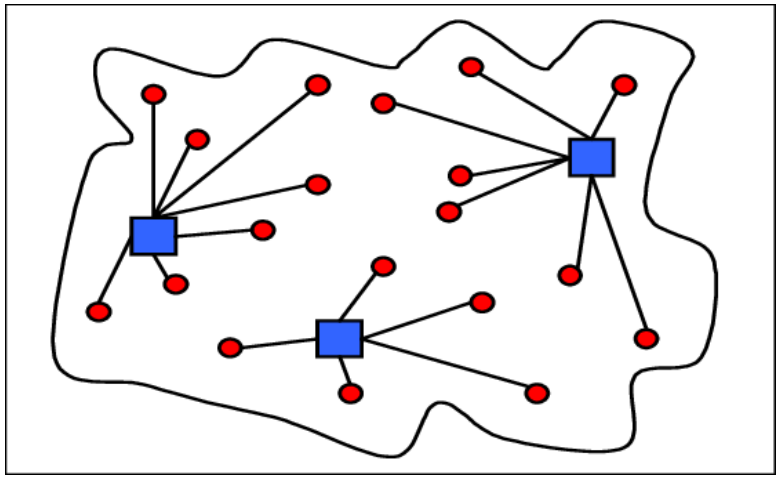
Minimum Norm Matroid Median

- Finite set $\mathcal{F} = \mathcal{D}$ of facilities and clients in a metric space
- A matroid $M = (\mathcal{F}, \mathcal{I})$
- For $F \in \mathcal{I}$, let $c \in \mathbb{R}_+^{\mathcal{D}}$ be such that c_j is the distance from $j \in \mathcal{D}$ to the nearest $i \in F$



Minimum Norm Matroid Median

- Finite set $\mathcal{F} = \mathcal{D}$ of facilities and clients in a metric space
- A matroid $M = (\mathcal{F}, \mathcal{I})$
- For $F \in \mathcal{I}$, let $c \in \mathbb{R}_+^{\mathcal{D}}$ be such that c_j is the distance from $j \in \mathcal{D}$ to the nearest $i \in F$
- Open facilities $F \in \mathcal{I}$ minimizing $f(c)$



Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)

k-center

Matroid median

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$

k-center

Matroid median

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

Matroid median

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

- 2-approx. by Hochbaum, Shmoys (optimal)

Matroid median

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

- 2-approx. by Hochbaum, Shmoys (optimal)

Matroid median

- 360-approx. by Krishnaswamy et al. (1st constant approx.)

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

- 2-approx. by Hochbaum, Shmoys (optimal)

Matroid median

- 360-approx. by Krishnaswamy et al. (1st constant approx.)
- 8-approx. by Swamy

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

- 2-approx. by Hochbaum, Shmoys (optimal)

Matroid median

- 360-approx. by Krishnaswamy et al. (1st constant approx.)
- 8-approx. by Swamy
- 7.081-approx. by Krishnaswamy et al.

Minimum norm k-median

Related Work

k-median

- $6\frac{2}{3}$-approx. by Charikar, Guha (1st constant approx.)
- Improvements: $6 \to 4 \to 3 + \epsilon \to 2.732$
- 2.611 by Byrka et al.

k-center

- 2-approx. by Hochbaum, Shmoys (optimal)

Matroid median

- 360-approx. by Krishnaswamy et al. (1st constant approx.)
- 8-approx. by Swamy
- 7.081-approx. by Krishnaswamy et al.

Minimum norm k-median

- $O(1)$-approx. by Chakrabarty and Swamy

Top-ℓ Norms

- How do we minimize a monotone, symmetric norm?

Top-ℓ Norms

- How do we minimize a monotone, symmetric norm?
- It suffices to control all Top-ℓ norms!

Top-ℓ Norms

- How do we minimize a monotone, symmetric norm?
- It suffices to control all Top-ℓ norms!

Theorem [**Chakrabarty, Swamy. '19; Hardy, Littlewood '34**]

Let $\alpha, \beta \in \mathbb{R}_+$. If $x, y \in \mathbb{R}_+^n$ satisfies $\mathrm{Top}_\ell(x) \leq \alpha \, \mathrm{Top}_\ell(y) + \beta$ for all $\ell \in [n]$, then $f(x) \leq \alpha f(y) + \beta f(1, 0, \ldots, 0)$.

Idea!

- Consider an LP relaxation of the discrete problem

Idea!

- Consider an LP relaxation of the discrete problem
- Express simultaneous optimization as budget constraints ($\mathrm{Top}_\ell \leq \mathrm{OPT}_\ell$)

Idea!

- Consider an LP relaxation of the discrete problem
- Express simultaneous optimization as budget constraints ($\mathrm{Top}_\ell \leq \mathrm{OPT}_\ell$)
- Need to linearize Top_ℓ

Idea!

- Consider an LP relaxation of the discrete problem
- Express simultaneous optimization as budget constraints ($\mathrm{Top}_\ell \leq \mathrm{OPT}_\ell$)
- Need to linearize Top_ℓ
- What is OPT_ℓ?

Idea!

- Consider an LP relaxation of the discrete problem
- Express simultaneous optimization as budget constraints ($\mathrm{Top}_\ell \leq \mathrm{OPT}_\ell$)
- Need to linearize Top_ℓ
- What is OPT_ℓ?

Proxy Function

- Let $z^+ := \max(z, 0)$

Proxy Function

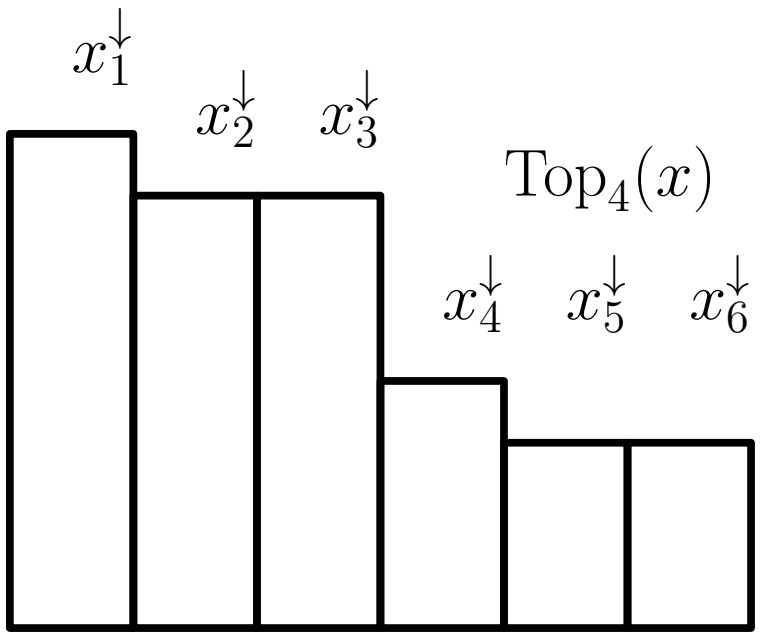
- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^{n} (x_i - \rho)^+$

Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^n (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$

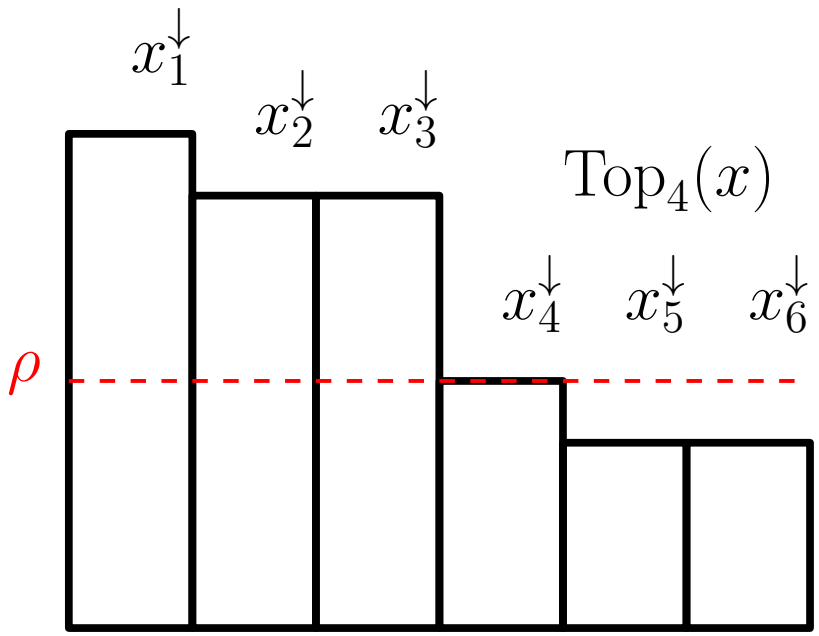
Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^n (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$



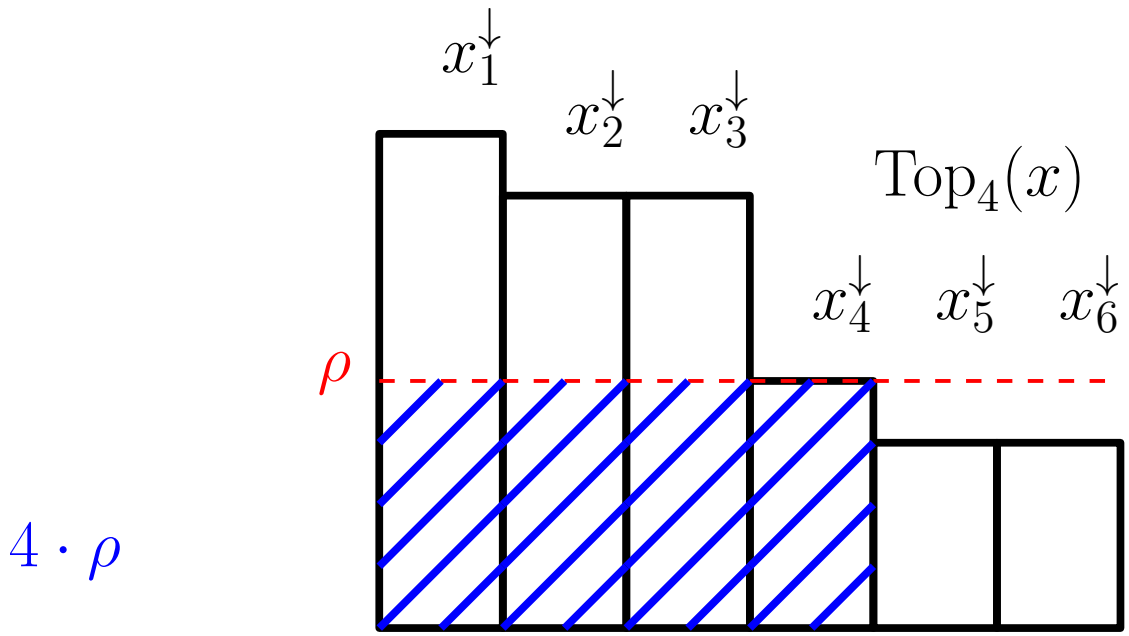
Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^n (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$



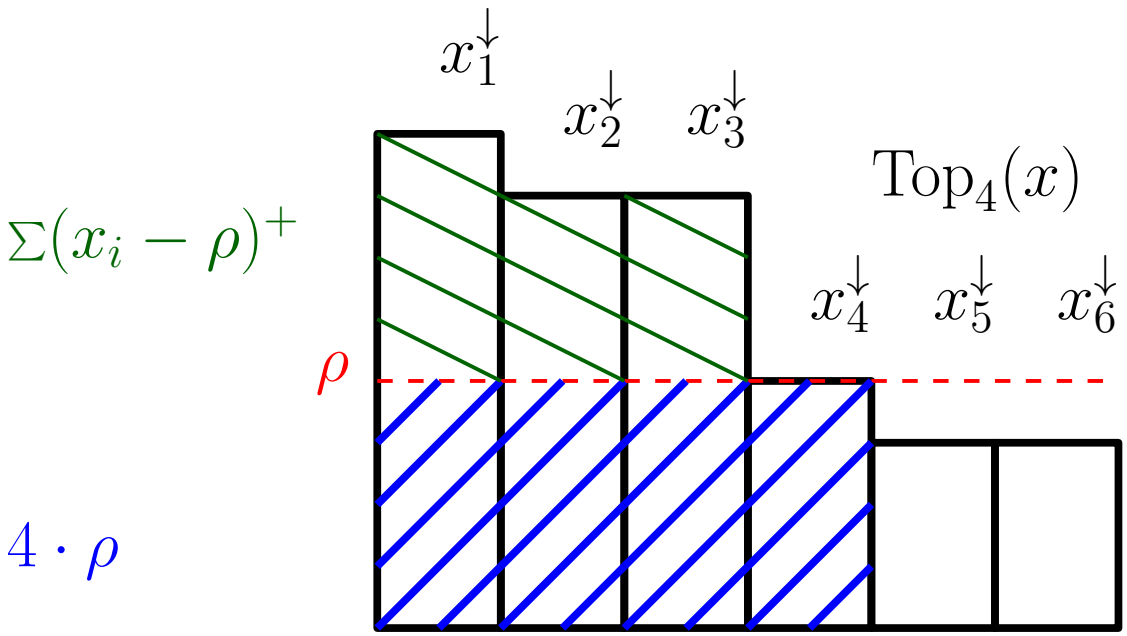
Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^{n} (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$



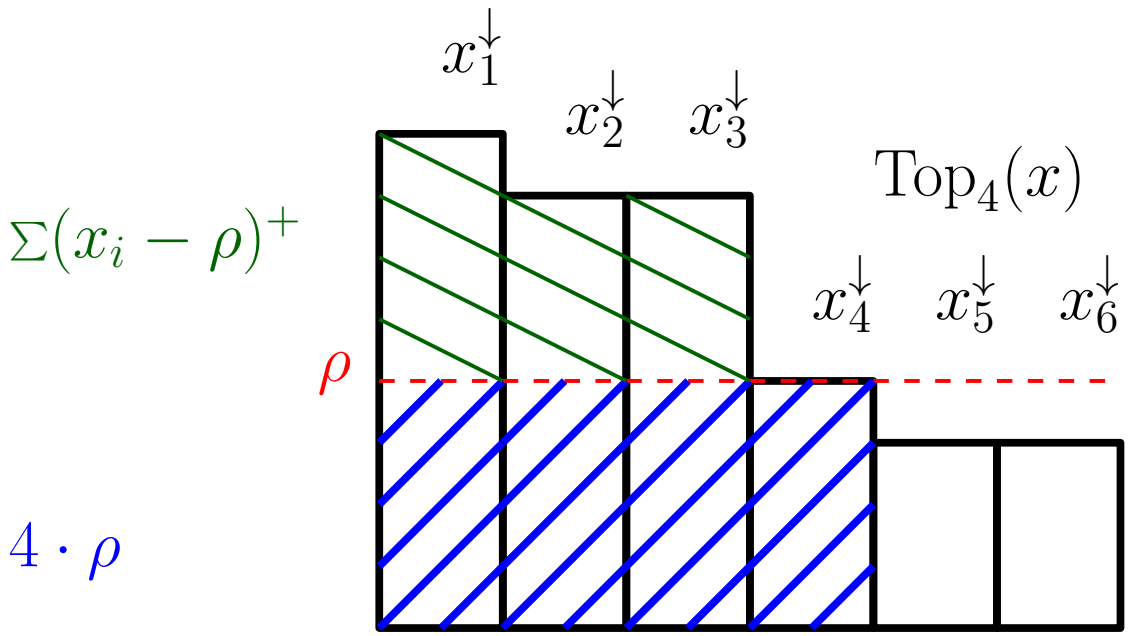
Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^{n} (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$



Proxy Function

- Let $z^+ := \max(z, 0)$
- $\mathrm{Top}_\ell(x) = \min_{\rho \geq 0} \ell\rho + \sum_{i=1}^{n} (x_i - \rho)^+$
- Equality attained when $\rho = x_\ell^\downarrow$
- What is the threshold ρ?



Sparsification

- Define POS $:= \{1, 2, 4, 8, \ldots, n\}$

Sparsification

- Define POS $:= \{1, 2, 4, 8, \ldots, n\}$
- If $\ell \in [2^m, 2^{m+1})$, then
  $\mathrm{Top}_{2^m}(x) \leq \mathrm{Top}_\ell(x) \leq \mathrm{Top}_{2^{m+1}}(x) \leq 2\,\mathrm{Top}_{2^m}(x)$

Sparsification

- Define POS $:= \{1, 2, 4, 8, \ldots, n\}$
- If $\ell \in [2^m, 2^{m+1})$, then
  $\mathrm{Top}_{2^m}(x) \leq \mathrm{Top}_\ell(x) \leq \mathrm{Top}_{2^{m+1}}(x) \leq 2\,\mathrm{Top}_{2^m}(x)$
- It suffices to control Top_ℓ norms for $\ell = 2^m, m \geq 0$

Enumeration

- Let o denote an optimal cost vector

Enumeration

- Let o denote an optimal cost vector
- Notice that $(o_\ell^{\downarrow})_{\ell \in \mathrm{POS}}, (\mathrm{Top}_\ell(o))_{\ell \in \mathrm{POS}}$ are non-increasing vectors

Enumeration

- Let o denote an optimal cost vector
- Notice that $(o_\ell^{\downarrow})_{\ell \in \mathrm{POS}}, (\mathrm{Top}_\ell(o))_{\ell \in \mathrm{POS}}$ are non-increasing vectors
- In polynomimal time, we can enumerate non-increasing sequences

Enumeration

- Let o denote an optimal cost vector
- Notice that $(o_\ell^\downarrow)_{\ell \in \mathrm{POS}}, (\mathrm{Top}_\ell(o))_{\ell \in \mathrm{POS}}$ are non-increasing vectors
- In polynomimal time, we can enumerate non-increasing sequences
- We can guess $t_\ell \in \left[o_\ell^\downarrow, (1+\epsilon)o_\ell^\downarrow\right)$ for $\ell \in \mathrm{POS}$

Enumeration

- Let o denote an optimal cost vector
- Notice that $(o_\ell^\downarrow)_{\ell \in \mathrm{POS}}, (\mathrm{Top}_\ell(o))_{\ell \in \mathrm{POS}}$ are non-increasing vectors
- In polynomimal time, we can enumerate non-increasing sequences
- We can guess $t_\ell \in \left[o_\ell^\downarrow, (1+\epsilon)o_\ell^\downarrow\right)$ for $\ell \in \mathrm{POS}$
- We can guess $\beta_\ell \in O(\mathrm{Top}_\ell(o))$ for $\ell \in \mathrm{POS}$

Idea!

- Consider an LP relaxation of the discrete problem
- Express simultaneous optimization as budget constraints ($\mathrm{Top}_\ell \leq \mathrm{OPT}_\ell$)
- Need to linearize Top_ℓ
- What is OPT_ℓ?

Overview

- Iterate through guesses t, β

Overview

- Iterate through guesses t, β
- Define an LP relaxation $\mathrm{LP}(t, \beta)$ and solve for a fractional solution

Overview

- Iterate through guesses t, β
- Define an LP relaxation $\mathrm{LP}(t, \beta)$ and solve for a fractional solution
- Round to an integral solution with at most constant violation of budget constraints

Overview

- Iterate through guesses t, β
- Define an LP relaxation $\mathrm{LP}(t, \beta)$ and solve for a fractional solution
- Round to an integral solution with at most constant violation of budget constraints
  - Consolidate clients

Overview

- Iterate through guesses t, β
- Define an LP relaxation $\mathrm{LP}(t, \beta)$ and solve for a fractional solution
- Round to an integral solution with at most constant violation of budget constraints
  - Consolidate clients
  - Secondary clustering

Overview

- Iterate through guesses t, β
- Define an LP relaxation $\mathrm{LP}(t, \beta)$ and solve for a fractional solution
- Round to an integral solution with at most constant violation of budget constraints
  - Consolidate clients
  - Secondary clustering
  - Auxiliary rounding LP to get integral solution

LP Relaxation

Let $r : 2^{\mathcal{F}} \to \mathbb{R}_+$ be the rank function of the matroid on $\mathcal{F}$.

LP Relaxation

Let $r : 2^{\mathcal{F}} \to \mathbb{R}_+$ be the rank function of the matroid on $\mathcal{F}$.

$$\begin{aligned} \sum_{i \in \mathcal{F}} x_{ij} &\geq 1 && \forall j \\ x_{ij} &\leq y_i && \forall i, j \\ x_{ij} &\geq 0 && \forall i, j \end{aligned}$$

LP Relaxation

Let $r : 2^{\mathcal{F}} \to \mathbb{R}_+$ be the rank function of the matroid on $\mathcal{F}$.

$$\sum_{i \in \mathcal{F}} x_{ij} \geq 1 \qquad \forall j$$

$$x_{ij} \leq y_i \qquad \forall i, j$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

$$y(S) \leq r(S) \qquad \forall S \subseteq \mathcal{F}$$

LP Relaxation

Let $r : 2^{\mathcal{F}} \to \mathbb{R}_+$ be the rank function of the matroid on $\mathcal{F}$.

$$\sum_{i \in \mathcal{F}} x_{ij} \geq 1 \qquad \forall j$$

$$x_{ij} \leq y_i \qquad \forall i, j$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

$$y(S) \leq r(S) \qquad \forall S \subseteq \mathcal{F}$$

$$\ell t_\ell + \sum_{j \in \mathcal{D}} \sum_{i \in \mathcal{F}} (c_{ij} - t_\ell)^+ x_{ij} \leq \beta_\ell \qquad \forall \ell \in \mathrm{POS}$$

LP Relaxation

Let $r : 2^{\mathcal{F}} \to \mathbb{R}_+$ be the rank function of the matroid on $\mathcal{F}$.

$$\sum_{i \in \mathcal{F}} x_{ij} \geq 1 \qquad \forall j$$

$$x_{ij} \leq y_i \qquad \forall i, j$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

$$y(S) \leq r(S) \qquad \forall S \subseteq \mathcal{F}$$

$$\ell t_\ell + \sum_{j \in \mathcal{D}} \sum_{i \in \mathcal{F}} (c_{ij} - t_\ell)^+ x_{ij} \leq \beta_\ell \qquad \forall \ell \in \mathrm{POS}$$

Let $\bar{x}, \bar{y}$ be an optimal solution

LP Rounding

- Follows similar steps for k-median by Charikar, Guha

LP Rounding

- Follows similar steps for k-median by Charikar, Guha
- Krishnaswamy et al. showed how to extend for matroid median

LP Rounding

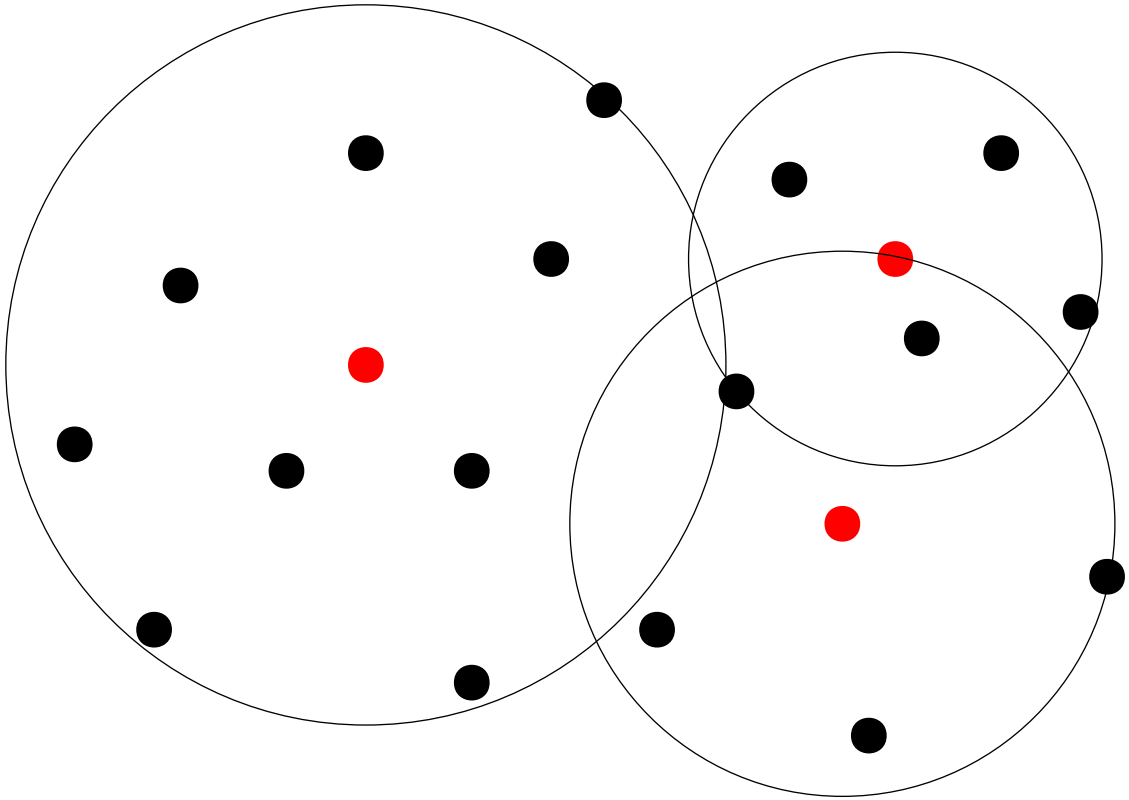
- Follows similar steps for k-median by Charikar, Guha
- Krishnaswamy et al. showed how to extend for matroid median
- Chakrabarty, Swamy showed how to extend for min-norm k-median

LP Rounding

- Follows similar steps for k-median by Charikar, Guha
- Krishnaswamy et al. showed how to extend for matroid median
- Chakrabarty, Swamy showed how to extend for min-norm k-median
- Highlight key ideas

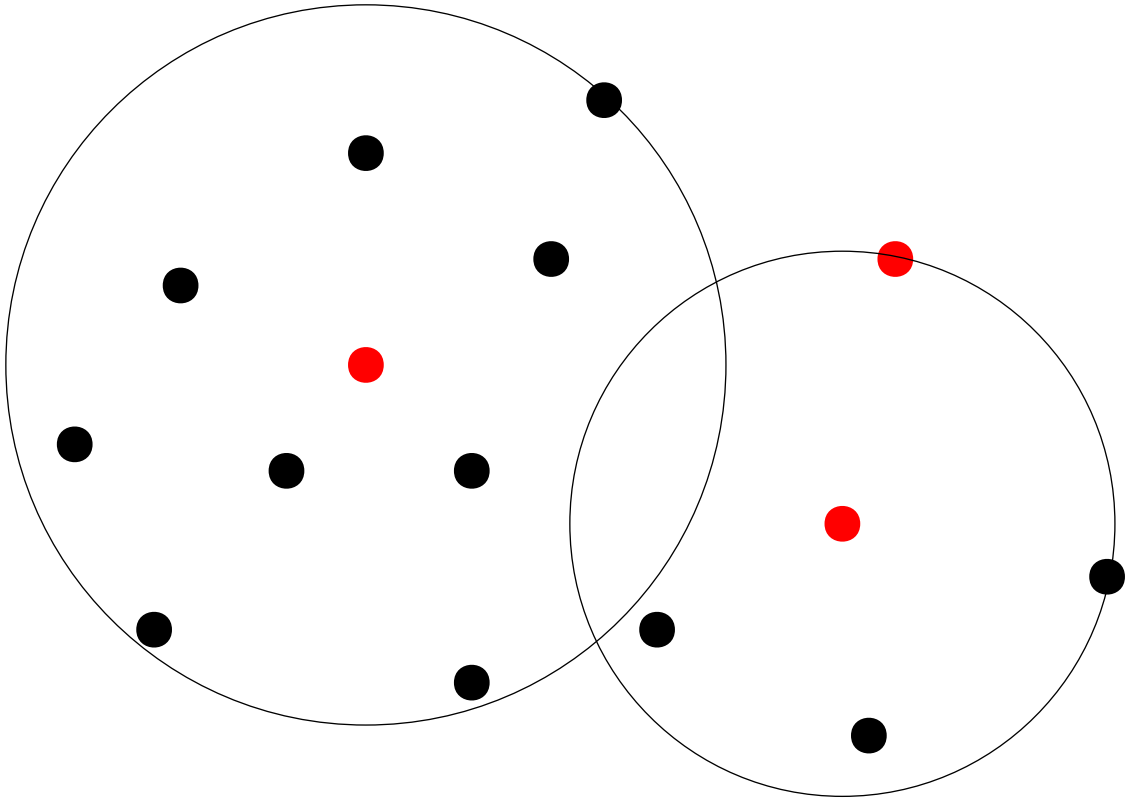
Demand Consolidation

"Combine clients who are close into one client"



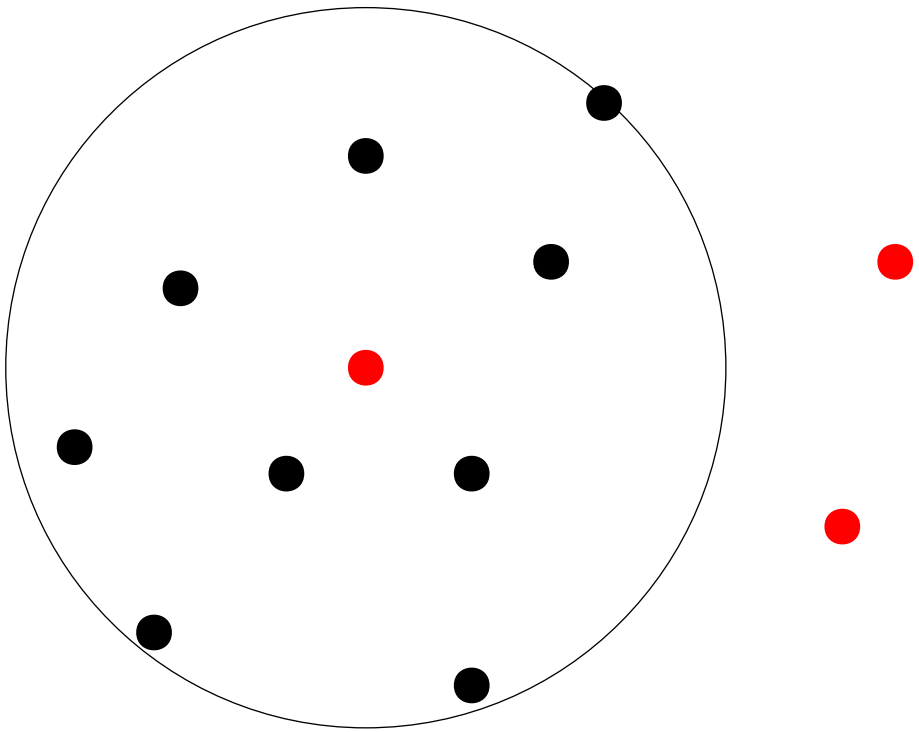
Demand Consolidation

"Combine clients who are close into one client"



Demand Consolidation

"Combine clients who are close into one client"

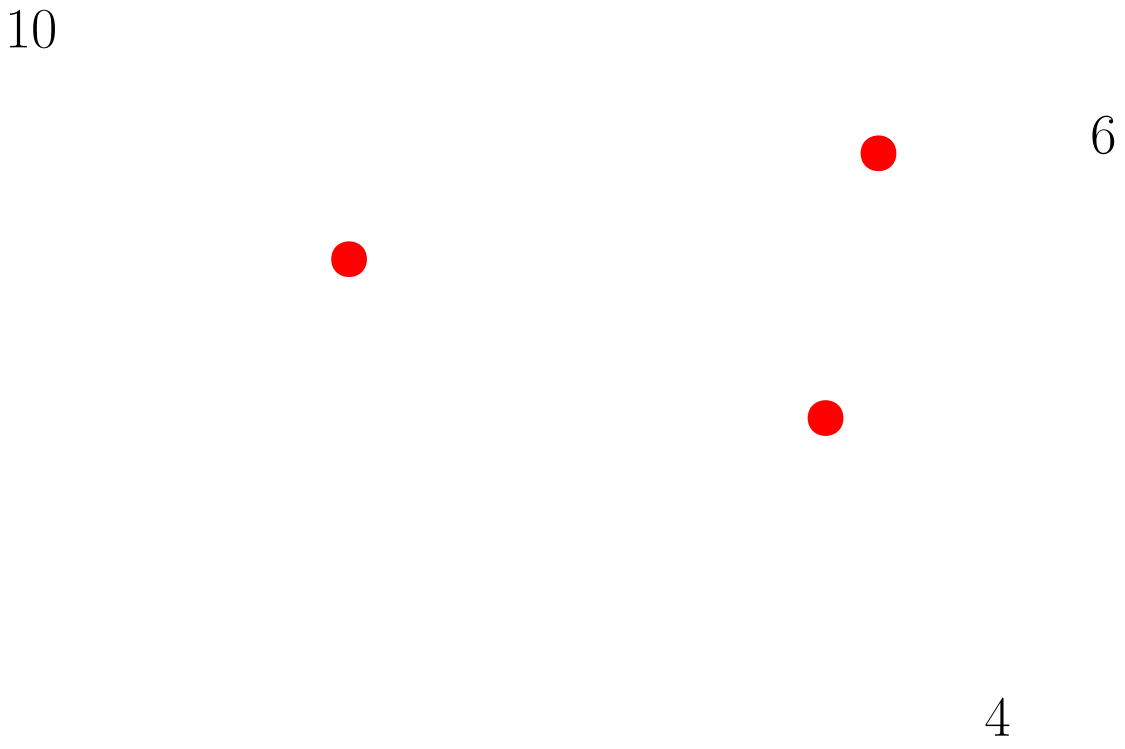


Demand Consolidation

"Combine clients who are close into one client"

Demand Consolidation

"Combine clients who are close into one client"



Secondary Clustering

- Consolidated clients are far apart

Secondary Clustering

- Consolidated clients are far apart
- There are disjoint balls B_j around consolidated client j in which $\sum_{i \in B_j} \bar{y}_i \geq \frac{1}{2}$

Secondary Clustering

- Consolidated clients are far apart
- There are disjoint balls B_j around consolidated client j in which $\sum_{i \in B_j} \bar{y}_i \geq \frac{1}{2}$
- Pair up nearby consolidated clients j, k so that $\sum_{i \in B_j \cup B_k} \bar{y}_i \geq 1$

Secondary Clustering

- Consolidated clients are far apart
- There are disjoint balls B_j around consolidated client j in which $\sum_{i \in B_j} \bar{y}_i \geq \frac{1}{2}$
- Pair up nearby consolidated clients j, k so that $\sum_{i \in B_j \cup B_k} \bar{y}_i \geq 1$
- Require at least one facility from $B_j \cup B_k$ to be open when rounding

Auxiliary LP for Facility Opening

$$
\begin{aligned}
y(S) &\leq r(S) && \forall S \subseteq \mathcal{F} \\
y(B_j) &\geq 1 && \forall j, \bar{y}(B_j) \geq 1 \\
y(B_j \cup B_k) &\geq 1 && \text{for all pairs } j, k \\
\text{Budget}_\ell(y) &\leq O(\beta_\ell) && \forall \ell \in \text{POS}
\end{aligned}
$$

Technical Roadblock

Fractional solution

- Independent in M

Technical Roadblock

Fractional solution

- Independent in M
- Basis in another partition matroid M'

Technical Roadblock

Fractional solution

- Independent in M
- Basis in another partition matroid M'
- Satisfies some budget constraints

Technical Roadblock

Fractional solution

- Independent in M
- Basis in another partition matroid M'
- Satisfies some budget constraints

"Round it to an integral independent set in M and basis in M' with at most constant violation of budget constraints."

Technical Roadblock

Fractional solution

- Independent in M
- Basis in another partition matroid M'
- Satisfies some budget constraints

"Round it to an integral independent set in M and basis in M' with at most constant violation of budget constraints."

In the case of perfect bipartite matching, the LP relaxation has a bad integrality gap with respect to violating Budget_ℓ constraints when expressed as cardinality constraints.

Technical Roadblock

- Our setup seems to have more structure than perfect bipartite matching

Technical Roadblock

- Our setup seems to have more structure than perfect bipartite matching
- We can get constant approximation of Top_ℓ matroid median and ordered matroid median using the objective function

Technical Roadblock

- Our setup seems to have more structure than perfect bipartite matching
- We can get constant approximation of Top_ℓ matroid median and ordered matroid median using the objective function
- May need further technical innovation

Thanks!